

Modeling the Economic Effects of the Section 232 Tariffs on Most Affected Industries

232 Model Release - 2021

Read Me :

This model accompanies the USITC report, *Economic Impact of Section 232 and 301 Tariffs on U.S. Industries*, Inv. 332-591. The report includes a quantitative analysis of the economic impact of section 232 tariffs on U.S. prices, production and imports. The model inputs (ORANGE-shaded cells) include elasticities that describe responsiveness to price changes, and domestic production and imports data. Data sources are listed in the cell above the data input. The model inputs include the effects of the 232 tariffs. Then, the model simulation estimates what prices and quantities would have looked like if section 232 tariffs were not in place. Outputs are in the GREEN-shaded cells. The white cells are intermediate calculations.

Note: NAICS 336300 is an aggregate of NAICS 336330 and NAICS 336340.

```
In[ ]:= ClearAll[f];  
Clear["Global`*"];
```

1. Policy Changes

Existing Section 232 tariff on aluminum imports

```
In[ ]:= t232a0 = 0.10;
```

Counterfactual Section 232 tariff on aluminum imports

```
In[ ]:= t232a1 = 0.00;
```

Existing Section 232 tariff on steel imports

```
In[ ]:= t232s0 = 0.25;
```

Counterfactual Section 232 tariff on steel imports

```
In[ ]:= t232s1 = 0.00;
```

2. Downstream Industry Definitions

Number of downstream industries

```
In[ ]:= i = 33;
```

Downstream industry NAICS code

```
Inames = {2110, 312110, 3149, 3322, 3323, 3324, 3325, 3326, 3327, 3328,
          3329, 3331, 3332, 3334, 3335, 3339, 3351, 3352, 3353, 3359, 336212, 336214,
          336350, 336370, 336390, 336300, 336411, 3365, 3369, 3372, 3399, 3336, 3366};

In[ ]:= Table[NAICSCodes[t] = Inames[[t]], {t, 1, i}];
```

3. Parameter Inputs

3.1 Elasticity of Substitution

Source: USITC's econometric estimation using the trade cost method in Riker (2020).

Riker, David. "A Trade Cost Approach to Estimating the Elasticity of Substitution." *Economics Working Paper Series 2020-07-D*, U.S. International Trade Commission, July 2020. https://www.usitc.gov/publications/332/working_papers/ecwp_2017-07-d.pdf.

Steel

```
In[ ]:= sigmas = 1.493;
```

Aluminum

```
In[ ]:= sigmaa = 6.046;
```

Downstream industries

```
In[ ]:= sigmad = {5.42, 1.67, 6.33, 9.54, 4.54, 3.84, 5.24, 5.51, 4.22,
                  5, 5.74, 6.95, 7.80, 9.27, 8.90, 7.02, 5.34, 6.67, 9.68, 7.82, 3.06,
                  4.83, 6.48, 4.02, 4.28, 6.44, 3, 7.12, 7.16, 4.27, 6.78, 8.30, 7.51};
```

3.2 Domestic Supply Elasticities

Source: USITC staff estimate based on recent AD/CVD staff reports and data on domestic capacity utilization.

USITC, *Cut-to-Length Carbon-Quality Steel Plate from India, Indonesia, Italy, Japan, and Korea*, December 2011; USITC, *Hot-Rolled Flat-Rolled Carbon-Quality Steel Products from Brazil, Japan, and Russia*, June 2011; USITC, *Certain Hot-Rolled Steel Flat Products from Australia, Brazil, Japan, Korea, the Netherlands, Turkey, and the United Kingdom*, July 2016; USITC, *Cold-Rolled Steel Flat Products from China and Japan*, July 2016; Bown and Russ, "Biden and Europe Remove Trump's Steel and Aluminum Tariffs, but It's Not Free Trade," November 11, 2021.

Domestic steel

```
In[ ]:= eds = 2.5;
```

Domestic aluminum

```
In[ ]:= eda = 4;
```

3.3 Import Supply Elasticities

Source: USITC staff estimate using pass-through results from chapter 6 econometric analysis.

Steel subject imports

```
ln[ ]:= eis = 15;
```

Steel non-subject imports

```
ln[ ]:= ens = 15;
```

Aluminum subject imports

```
ln[ ]:= eia = 20;
```

Aluminum non-subject imports

```
ln[ ]:= ena = 20;
```

4. Data Inputs

4.1 Shipments of U.S. Domestic Steel to Each Downstream Industry

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures. The use of steel by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report.

U.S. Census Bureau, "Annual Survey of Manufactures," accessed October 17, 2022. BEA, "Input-Output Accounts Data," accessed October 17, 2022.

```
ln[ ]:= vds0 = {5 120 026, 0, 505 548, 504 490, 7 767 193, 2 399 146, 219 065, 2 080 338, 3 254 856,
1 578 768, 2 747 835, 3 972 583, 675 302, 863 283, 872 552, 3 607 028, 416 553, 619 721,
820 113, 861 382, 191 524, 247 128, 860 352, 3 728 904, 1 881 696, 787 495,
925 304, 538 929, 762 770, 898 215, 739 778, 617 820, 218 138}; (*in thousands*)
```

```
ln[ ]:= pds0 = 1;
```

```
ln[ ]:= vdstot0 = Total[vds0];
```

```
ln[ ]:= Table[qds0[t] = (vds0[[t]] / pds0), {t, 1, i}];
```

```
ln[ ]:= qdstot0 = vdstot0 / pds0;
```

4.2 Shipments of U.S. Domestic Aluminum to Each Downstream Industry

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures. The use of aluminum by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values,

see Appendix F of the report.

U.S. Census Bureau, “Annual Survey of Manufactures,” accessed October 17, 2022. BEA, “Input-Output Accounts Data,” accessed October 17, 2022.

```
ln[ ]:= vda0 = {0, 5 563 109, 0, 242 467, 2 140 957, 4 897 695, 124 011, 36 321,
            1 104 757, 29 459, 1 384 429, 210 340, 248 357, 573 641, 400 861, 848 940,
            291 750, 143 952, 189 608, 579 807, 789 846, 129 390, 1 025 570, 1 920 951,
            969 358, 405 679, 46 960, 52 946, 455 975, 260 953, 259 003, 671 513, 523 199};
(*in thousands*)
```

```
ln[ ]:= pda0 = 1;
```

```
ln[ ]:= vdatot0 = Total[vda0];
```

```
ln[ ]:= Table[qda0[t] = vda0[[t]] / pda0, {t, 1, i}];
```

```
ln[ ]:= qdatot0 = vdatot0 / pda0;
```

4.3 Shipments of Steel Imports to Each Downstream Industry

Source: USITC estimates. Total import data were obtained from the USITC’s DataWeb. The use of steel by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report. USITC, “DataWeb/Census,” accessed September 28, 2022. BEA, “Input-Output Accounts Data,” accessed October 17, 2022.

Share of total steel imports that are subject to the Section 232 tariffs

```
ln[ ]:= shrs = 0.3149;
```

Subject imports

```
ln[ ]:= vis0 = shrs * {879 895, 0, 26 882, 310 973, 2 799 193, 810 883, 92 974, 363 562, 1 028 931,
                    988 140, 1 615 870, 2 279 952, 236 486, 290 791, 448 278, 1 823 940, 98 217, 227 513,
                    355 300, 295 185, 98 332, 123 397, 458 582, 1 272 406, 784 867, 330 501, 455 990,
                    227 513, 359 587, 339 342, 328 137, 344 162, 129 181}; (*in thousands*)
```

```
ln[ ]:= pis0 = 1;
```

```
ln[ ]:= vistot0 = Total[vis0];
```

```
ln[ ]:= Table[qis0[t] = vis0[[t]] / (pis0 (1 + t232s0)), {t, 1, i}];
```

```
ln[ ]:= qistot0 = vistot0 / (pis0 (1 + t232s0));
```

Non-subject imports

```
ln[ ]:= vns0 = (1 - shrs) * {879 895, 0, 26 882, 310 973, 2 799 193, 810 883, 92 974, 363 562,
                            1 028 931, 988 140, 1 615 870, 2 279 952, 236 486, 290 791, 448 278, 1 823 940, 98 217,
                            227 513, 355 300, 295 185, 98 332, 123 397, 458 582, 1 272 406, 784 867, 330 501,
                            455 990, 227 513, 359 587, 339 342, 328 137, 344 162, 129 181}; (*in thousands*)
```

```
ln[ ]:= pns0 = 1;
```

```
ln[ ]:= vnstot0 = Total[vns0];

ln[ ]:= Table[qns0[t] = vns0[[t]] / (pns0), {t, 1, i}];

ln[ ]:= qnstot0 = vnstot0 / (pns0);
```

4.4 Shipments of Aluminum Imports to Each Downstream Industry

Source: USITC estimates. Total import data were obtained from the USITC's DataWeb. The use of aluminum by downstream industry was calculated using the Bureau of Economic Analysis (BEA) 2012 Use Table and 2012 Import Matrix with adjustments by industry analysts as necessary. For more information on the steps taken to calculate these values, see Appendix F of the report. USITC, "DataWeb/Census," accessed September 28, 2022. BEA, "Input-Output Accounts Data," accessed October 17, 2022.

Share of total aluminum imports that are subject to the Section 232 tariffs

```
ln[ ]:= shra = 0.3099;
```

Subject imports

```
ln[ ]:= via0 = shra * {0, 1715716, 0, 117700, 666162, 2337410, 31419,
17970, 355666, 6400, 457676, 1954, 72971, 201984, 91442, 328078, 80257,
61600, 55879, 197109, 247438, 190855, 88040, 655482, 404325, 170258,
13317, 0, 271643, 145212, 99951, 33202, 151806}; (*in thousands*)
```

```
ln[ ]:= pia0 = 1;
```

```
ln[ ]:= viatot0 = Total[via0];
```

```
ln[ ]:= Table[qia0[t] = via0[[t]] / (pia0 (1 + t232a0)), {t, 1, i}];
```

```
ln[ ]:= qiatot0 = viatot0 / (pia0 (1 + t232a0));
```

Non-subject imports

```
ln[ ]:= vna0 = (1 - shra) * {0, 1715716, 0, 117700, 666162, 2337410, 31419,
17970, 355666, 6400, 457676, 1954, 72971, 201984, 91442, 328078, 80257,
61600, 55879, 197109, 247438, 190855, 88040, 655482, 404325, 170258,
13317, 0, 271643, 145212, 99951, 33202, 151806}; (*in thousands*)
```

```
ln[ ]:= pna0 = 1;
```

```
ln[ ]:= vnatot0 = Total[vna0];
```

```
ln[ ]:= Table[qna0[t] = vna0[[t]] / (pna0), {t, 1, i}];
```

```
ln[ ]:= qnatot0 = vnatot0 / (pna0);
```

4.5 Downstream data

Source: USITC estimates. Total domestic production data were obtained from the U.S. Census Annual Survey of Manufactures or estimated by the Commission's industry analysts using available production data from industry sources, if not available from the ASM. U.S. imports and exports data were obtained from USITC's DataWeb.

U.S. Census Bureau, "Annual Survey of Manufactures," accessed October 17, 2022. USITC, "DataWeb/Census," accessed September

28, 2022.

Downstream domestic production

```
ln["v"]:= vdo = {429 000 000, 44 416 236, 9 694 852, 9 724 369, 96 325 734,  
35 519 032, 8 694 931, 9 023 239, 62 286 800, 24 710 245, 73 832 139, 83 586 904,  
36 069 430, 45 787 347, 28 191 665, 101 859 161, 12 128 140, 25 137 430, 40 154 354,  
50 921 581, 9 899 698, 19 110 844, 36 752 028, 30 075 436, 54 777 590, 21 700 684,  
88 765 120, 9 163 065, 18 920 621, 23 952 769, 58 782 675, 36 499 826, 38 418 677};  
(*in thousands*)
```

Downstream domestic exports

```
ln[*:]= vde0 = {134 357 186, 1 319 822, 1 433 103, 2 495 184, 1 868 945, 3 776 788,  
2 653 680, 1 528 682, 3 679 256, 0, 22 865 398, 23 254 669, 30 543 414, 6 433 768,  
5 245 827, 30 311 794, 1 518 251, 7 695 752, 13 814 849, 20 425 334, 1 289 020,  
2 134 359, 2 855 712, 1 102 580, 11 131 932, 11 061 133, 2 976 752, 1 988 882,  
3 221 714, 1 225 442, 16 017 549, 21 571 787, 2 101 801}; (*in thousands*)
```

```
ln[•]:= Table[vdd0[t] = vd0[t] - vde0[t], {t, 1, i}]
```

$Out_{\neq} = \{294\,642\,814, 43\,096\,414, 8\,261\,749, 7\,229\,185, 94\,456\,789, 31\,742\,244, 6\,041\,251, 7\,494\,557, 58\,607\,544, 24\,710\,245, 50\,966\,741, 60\,332\,235, 5\,526\,016, 39\,353\,579, 22\,945\,838, 71\,547\,367, 10\,609\,889, 17\,441\,678, 26\,339\,505, 30\,496\,247, 8\,610\,678, 16\,976\,485, 33\,896\,316, 28\,972\,856, 43\,645\,658, 10\,639\,551, 85\,788\,368, 7\,174\,183, 15\,698\,907, 22\,727\,327, 42\,765\,126, 14\,928\,039, 36\,316\,876\}$

Downstream imports

```
ln["v"]:= vi0 = {138046390, 4241965, 10076848, 11856648, 8325915, 6010580, 9049323, 3822849,
6379728, 0, 40699817, 40178731, 28623222, 21755813, 14155970, 63361431, 15399471,
43037216, 38753335, 54978192, 2580465, 1011606, 9448766, 685853, 40219293,
21898756, 13104819, 1095840, 10645573, 8193422, 112486883, 25184123, 3404510};
(*in thousands*)
```

```
ln[*]:= pi0 = {1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1};
```

```
ln[6]:= Table[qi0[t] = vi0[[t]] / (pi0[[t]]), {t, 1, i}] ;
```

4.6 Calculated Cost Shares by Downstream Industry

Cost share of aluminum in production of each downstream product

```
lnf:= N[Table[csa[t] = (vda0[[t]] + via0[[t]] + vna0[[t]]) / vd0[[t]], {t, 1, i}]]
```

```
Out[5]= {0., 0.163878, 0., 0.0370376, 0.0291419, 0.203697, 0.0178759, 0.0060168, 0.0234467,
0.00145118, 0.0249499, 0.0025398, 0.0089086, 0.0169397, 0.0174627, 0.0115553, 0.030673,
0.00817713, 0.00611358, 0.0152571, 0.104779, 0.0167572, 0.0303006, 0.0856657, 0.0250775,
0.02654, 0.000679062, 0.0057782, 0.0384563, 0.0169569, 0.00610646, 0.0193074, 0.0175697}
```


5. Parameter Calibration

5.1 Supply parameters

$$\ln[\circ] := \text{ads} = \text{qdstot0 pds0}^{-\text{eds}};$$

$$\ln[\circ] := \text{ais} = \text{qistot0 pis0}^{-\text{eis}};$$

$$\ln[\circ] := \text{ans} = \text{qnstot0 pns0}^{-\text{ens}};$$

$$\ln[\circ] := \text{ada} = \text{qdatot0 pda0}^{-\text{eda}};$$

$$\ln[\circ] := \text{aia} = \text{qiatot0 pia0}^{-\text{eia}};$$

$$\ln[\circ] := \text{ana} = \text{qnatot0 pna0}^{-\text{ena}};$$

5.1 Demand parameters

$$\ln[\circ] := \text{Table}[\text{bia}[t] = \text{If}[\text{vda0}[[t]] > 0, \left(\frac{\text{pda0}}{\text{pia0} (1 + t232a0)} \right)^{1-\text{sigmaa}} \left(\frac{\text{via0}[[t]]}{\text{vda0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{bna}[t] = \text{If}[\text{vda0}[[t]] > 0, \left(\frac{\text{pda0}}{\text{pna0}} \right)^{1-\text{sigmaa}} \left(\frac{\text{vna0}[[t]]}{\text{vda0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{za0}[t] = \left(\text{pda0}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia0} (1 + t232a0))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna0}^{1-\text{sigmaa}} \right)^{\frac{1}{1-\text{sigmaa}}}, \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{bis}[t] = \text{If}[\text{vds0}[[t]] > 0, \left(\frac{\text{pds0}}{\text{pis0} (1 + t232s0)} \right)^{1-\text{sigmas}} \left(\frac{\text{vis0}[[t]]}{\text{vds0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{bns}[t] = \text{If}[\text{vds0}[[t]] > 0, \left(\frac{\text{pds0}}{\text{pns0}} \right)^{1-\text{sigmas}} \left(\frac{\text{vns0}[[t]]}{\text{vds0}[[t]]} \right), 0], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{zs0}[t] = \left(\text{pds0}^{1-\text{sigmas}} + \text{bis}[t] (\text{pis0} (1 + t232s0))^{1-\text{sigmas}} + \text{bns}[t] \text{pns0}^{1-\text{sigmas}} \right)^{\frac{1}{1-\text{sigmas}}}, \{t, 1, i\}];$$

Calibration of cost parameters

$$\ln[\circ] := \text{Table}[\text{pd0}[t] = \text{w0}[[t]] + \text{bs0}[t] \times \text{zs0}[t] + \text{ba0}[t] \times \text{za0}[t], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{qd0}[t] = \text{vdd0}[t] / \text{pd0}[t], \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{bi}[t] = \left(\frac{\text{pd0}[t]}{\text{pi0}[[t]]} \right)^{1-\text{sigmad}[[t]]} \left(\frac{\text{vi0}[[t]]}{\text{vdd0}[t]} \right), \{t, 1, i\}];$$

$$\ln[\circ] := \text{Table}[\text{P0}[t] = \left(\text{pd0}[t]^{1-\text{sigmad}[[t]]} + \text{bi}[t] \text{pi0}[[t]]^{1-\text{sigmad}[[t]]} \right)^{\frac{1}{1-\text{sigmad}[[t]]}}, \{t, 1, i\}];$$

$ln[] := \text{Table}[Cs[t] = css[t] ==$

$$\left(\frac{1}{vd0[t]} \right) \left(\left(pds0 \left(\frac{k0[t] \times bs0[t]}{pds0} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pds0}{zs0[t]} \right)^{1-sigmas} \right) + \right. \\ \left. \left(pis0 (1 + t232s0) \left(\frac{k0[t] \times bs0[t] \times bis[t]}{pis0 (1 + t232s0)} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \right. \right. \\ \left. \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pis0 (1 + t232s0)}{zs0[t]} \right)^{1-sigmas} \right) + \left(pns0 \left(\frac{k0[t] \times bs0[t] \times bns[t]}{pns0} \right) \right. \\ \left. \left. \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{zs0[t]}{pd0[t]} \right) \left(\frac{pns0}{zs0[t]} \right)^{1-sigmas} \right) \right), \{t, 1, i\}];$$

$ln[] := \text{Table}[Ca[t] =$

$$csa[t] == \left(\frac{1}{vd0[t]} \right) \left(\left(pda0 \left(\frac{k0[t] \times ba0[t]}{pda0} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pda0}{za0[t]} \right)^{1-sigmaa} \right) + \right. \\ \left. \left(pia0 (1 + t232a0) \left(\frac{k0[t] \times ba0[t] \times bia[t]}{pia0 (1 + t232a0)} \right) \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \right. \right. \\ \left. \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pia0 (1 + t232a0)}{za0[t]} \right)^{1-sigmaa} \right) + \left(pna0 \left(\frac{k0[t] \times ba0[t] \times bna[t]}{pna0} \right) \right. \\ \left. \left. \left(\frac{pd0[t]}{p0[t]} \right)^{1-sigmad[t]} \left(\frac{za0[t]}{pd0[t]} \right) \left(\frac{pna0}{za0[t]} \right)^{1-sigmaa} \right) \right), \{t, 1, i\}];$$

$ln[] := \text{Table}[Ck[t] = k0[t] == p0[t]^{1-sigmad[t]} pd0[t]^{sigmad[t]} qd0[t], \{t, 1, i\}];$

```

In[ ]:= sol = FindRoot[{Cs[1], Ca[1], Cs[2], Ca[2], Cs[3], Ca[3], Cs[4], Ca[4], Cs[5], Ca[5],
Cs[6], Ca[6], Cs[7], Ca[7], Cs[8], Ca[8], Cs[9], Ca[9], Cs[10], Ca[10], Cs[11],
Ca[11], Cs[12], Ca[12], Cs[13], Ca[13], Cs[14], Ca[14], Cs[15], Ca[15], Cs[16],
Ca[16], Cs[17], Ca[17], Cs[18], Ca[18], Cs[19], Ca[19], Cs[20], Ca[20], Cs[21],
Ca[21], Cs[22], Ca[22], Cs[23], Ca[23], Cs[24], Ca[24], Cs[25], Ca[25], Cs[26],
Ca[26], Cs[27], Ca[27], Cs[28], Ca[28], Cs[29], Ca[29], Cs[30], Ca[30], Cs[31],
Ca[31], Cs[32], Ca[32], Cs[33], Ca[33], Ck[1], Ck[2], Ck[3], Ck[4], Ck[5],
Ck[6], Ck[7], Ck[8], Ck[9], Ck[10], Ck[11], Ck[12], Ck[13], Ck[14], Ck[15],
Ck[16], Ck[17], Ck[18], Ck[19], Ck[20], Ck[21], Ck[22], Ck[23], Ck[24],
Ck[25], Ck[26], Ck[27], Ck[28], Ck[29], Ck[30], Ck[31], Ck[32], Ck[33]},
{bs0[1], css[1]}, {ba0[1], csa[1]}, {k0[1], vdd0[1] + vi0[1]}, {bs0[2], css[2]},
{ba0[2], csa[2]}, {k0[2], vdd0[2] + vi0[2]}, {bs0[3], css[3]},
{ba0[3], csa[3]}, {k0[3], vdd0[3] + vi0[3]}, {bs0[4], css[4]},
{ba0[4], csa[4]}, {k0[4], vdd0[4] + vi0[4]}, {bs0[5], css[5]},
{ba0[5], csa[5]}, {k0[5], vdd0[5] + vi0[5]}, {bs0[6], css[6]},
{ba0[6], csa[6]}, {k0[6], vdd0[6] + vi0[6]}, {bs0[7], css[7]},
{ba0[7], csa[7]}, {k0[7], vdd0[7] + vi0[7]}, {bs0[8], css[8]}, {ba0[8], csa[8]},
{k0[8], vdd0[8] + vi0[8]}, {bs0[9], css[9]}, {ba0[9], csa[9]}, {k0[9], vdd0[9] + vi0[9]},
{bs0[10], css[10]}, {ba0[10], csa[10]}, {k0[10], vdd0[10] + vi0[10]},
{bs0[11], css[11]}, {ba0[11], csa[11]}, {k0[11], vdd0[11] + vi0[11]},
{bs0[12], css[12]}, {ba0[12], csa[12]}, {k0[12], vdd0[12] + vi0[12]},
{bs0[13], css[13]}, {ba0[13], csa[13]}, {k0[13], vdd0[13] + vi0[13]},
{bs0[14], css[14]}, {ba0[14], csa[14]}, {k0[14], vdd0[14] + vi0[14]},
{bs0[15], css[15]}, {ba0[15], csa[15]}, {k0[15], vdd0[15] + vi0[15]},
{bs0[16], css[16]}, {ba0[16], csa[16]}, {k0[16], vdd0[16] + vi0[16]},
{bs0[17], css[17]}, {ba0[17], csa[17]}, {k0[17], vdd0[17] + vi0[17]},
{bs0[18], css[18]}, {ba0[18], csa[18]}, {k0[18], vdd0[18] + vi0[18]},
{bs0[19], css[19]}, {ba0[19], csa[19]}, {k0[19], vdd0[19] + vi0[19]},
{bs0[20], css[20]}, {ba0[20], csa[20]}, {k0[20], vdd0[20] + vi0[20]},
{bs0[21], css[21]}, {ba0[21], csa[21]}, {k0[21], vdd0[21] + vi0[21]},
{bs0[22], css[22]}, {ba0[22], csa[22]}, {k0[22], vdd0[22] + vi0[22]},
{bs0[23], css[23]}, {ba0[23], csa[23]}, {k0[23], vdd0[23] + vi0[23]},
{bs0[24], css[24]}, {ba0[24], csa[24]}, {k0[24], vdd0[24] + vi0[24]},
{bs0[25], css[25]}, {ba0[25], csa[25]}, {k0[25], vdd0[25] + vi0[25]},
{bs0[26], css[26]}, {ba0[26], csa[26]}, {k0[26], vdd0[26] + vi0[26]},
{bs0[27], css[27]}, {ba0[27], csa[27]}, {k0[27], vdd0[27] + vi0[27]},
{bs0[28], css[28]}, {ba0[28], csa[28]}, {k0[28], vdd0[28] + vi0[28]},
{bs0[29], css[29]}, {ba0[29], csa[29]}, {k0[29], vdd0[29] + vi0[29]},
{bs0[30], css[30]}, {ba0[30], csa[30]}, {k0[30], vdd0[30] + vi0[30]},
{bs0[31], css[31]}, {ba0[31], csa[31]}, {k0[31], vdd0[31] + vi0[31]},
{bs0[32], css[32]}, {ba0[32], csa[32]}, {k0[32], vdd0[32] + vi0[32]},
{bs0[33], css[33]}, {ba0[33], csa[33]}, {k0[33], vdd0[33] + vi0[33]}}];

```

$\text{In}[*]:= \text{Table}[\text{bs}[t] = \text{bs0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{0.0286739, 0., 0.0765174, 0.356796, 0.243285, 0.272072, 0.114734, 0.678133, 0.141466, 0.311207, 0.249109, 0.291266, 0.390515, 0.0555682, 0.144916, 0.19181, 0.0813401, 0.0974821, 0.0978902, 0.073613, 0.0922292, 0.0517365, 0.0997184, 0.424029, 0.136553, 0.254411, 0.0369115, 0.246386, 0.1774, 0.112446, 0.054398, 0.178094, 0.0252785\}$

$\text{In}[*]:= \text{Table}[\text{ba}[t] = \text{ba0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{0., 0.214339, 0., 0.0643503, 0.0365297, 0.367034, 0.0291622, 0.0117671, 0.0291975, 0.00168647, 0.043551, 0.00394828, 0.0787713, 0.0220022, 0.024264, 0.0193374, 0.0401478, 0.0134596, 0.0103689, 0.0288244, 0.150318, 0.0235328, 0.0359758, 0.127636, 0.0371652, 0.0690122, 0.000750879, 0.00833166, 0.0576381, 0.0210298, 0.0092635, 0.0536512, 0.0201153\}$

$\text{In}[*]:= \text{Table}[\text{k}[t] = \text{k0}[t] /. \text{sol}, \{t, 1, i\}]$

$\text{Out}[*]= \{4.32689 \times 10^8, 4.73384 \times 10^7, 1.83386 \times 10^7, 1.90858 \times 10^7, 1.02783 \times 10^8, 3.77528 \times 10^7, 1.50906 \times 10^7, 1.13174 \times 10^7, 6.49873 \times 10^7, 2.47102 \times 10^7, 9.16666 \times 10^7, 1.00511 \times 10^8, 3.41492 \times 10^7, 6.11094 \times 10^7, 3.71018 \times 10^7, 1.34909 \times 10^8, 2.60094 \times 10^7, 6.04789 \times 10^7, 6.50928 \times 10^7, 8.54744 \times 10^7, 1.11911 \times 10^7, 1.79881 \times 10^7, 4.33451 \times 10^7, 2.96587 \times 10^7, 8.3865 \times 10^7, 3.25383 \times 10^7, 9.88932 \times 10^7, 8.27002 \times 10^6, 2.63445 \times 10^7, 3.09207 \times 10^7, 1.55252 \times 10^8, 4.01122 \times 10^7, 3.97214 \times 10^7\}$

$\text{In}[*]:= \text{Table}[\text{pd0}[t] = (\text{w0}[t] + \text{bs}[t] \times \text{zs0}[t] + \text{ba}[t] \times \text{za0}[t]), \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{qd0}[t] = \text{vdd0}[t] / \text{pd0}[t], \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{bi}[t] = \left(\frac{\text{pd0}[t]}{\text{pi0}[t]} \right)^{1-\text{sigmad}[t]} \left(\frac{\text{vi0}[t]}{\text{vdd0}[t]} \right), \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{Pindex0}[t] = (\text{pd0}[t]^{1-\text{sigmad}[t]} + \text{bi}[t] \text{pi0}[t]^{1-\text{sigmad}[t]})^{\frac{1}{1-\text{sigmad}[t]}}, \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{za0}[t] = \text{If}[\text{adinp}[t] + \text{aiinp}[t] + \text{aninp}[t] > 0, (\text{pda0}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia0} (1 + \text{t232a0}))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna0}^{1-\text{sigmaa}})^{\frac{1}{1-\text{sigmaa}}}, 0], \{t, 1, i\}];$

$\text{In}[*]:= \text{Table}[\text{zs0}[t] = \text{If}[\text{sdinp}[t] + \text{siinp}[t] + \text{sninp}[t] > 0, (\text{pds0}^{1-\text{sigmas}} + \text{bis}[t] (\text{pis0} (1 + \text{t232s0}))^{1-\text{sigmas}} + \text{bns}[t] \text{pns0}^{1-\text{sigmas}})^{\frac{1}{1-\text{sigmas}}}, 0], \{t, 1, i\}];$

6. New Equilibrium

$\text{In}[*]:= \text{w} = \text{w0};$

$\text{In}[*]:= \text{pi} = \text{pi0};$

$\text{In}[*]:= \text{Table}[\text{za}[t] = (\text{pda}^{1-\text{sigmaa}} + \text{bia}[t] (\text{pia} (1 + \text{t232a1}))^{1-\text{sigmaa}} + \text{bna}[t] \text{pna}^{1-\text{sigmaa}})^{\frac{1}{1-\text{sigmaa}}}, \{t, 1, i\}];$

```
ln[ ]:= Table[
  zs[t] = (pds1-sigmas + bis[t] (pis (1 + t232s1))1-sigmas + bns[t] pns1-sigmas) $\frac{1}{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Table[pd[t] = (w[[t]] + bs[t] × zs[t] + ba[t] × za[t]), {t, 1, i}];
```

```
ln[ ]:= Table[P[t] = (pd[t]1-sigmad[[t]] + bi[t] pi[[t]]1-sigmad[[t]]) $\frac{1}{1-sigmad[[t]]}$ , {t, 1, i}];
```

Equilibrium equations

```
ln[ ]:= Table[qda[t] =  $\left(\frac{k[t] \times ba[t]}{pda}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pda}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Eda = ada pdaeda == Sum[qda[t], {t, 1, i}];
```

```
ln[ ]:= Table[qia[t] =
   $\left(\frac{k[t] \times ba[t] \times bia[t]}{pia (1 + t232a1)}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pia (1 + t232a1)}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Eia = aia piaeia == Sum[qia[t], {t, 1, i}];
```

```
ln[ ]:= Table[qna[t] =  $\left(\frac{k[t] \times ba[t] \times bna[t]}{pna}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{za[t]}{pd[t]}\right) \left(\frac{pna}{za[t]}\right)^{1-sigmaa}$ , {t, 1, i}];
```

```
ln[ ]:= Ena = ana pnaena == Sum[qna[t], {t, 1, i}];
```

```
ln[ ]:= Table[qds[t] =  $\left(\frac{k[t] \times bs[t]}{pds}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pds}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Eds = ads pdseds == Sum[qds[t], {t, 1, i}];
```

```
ln[ ]:= Table[qis[t] =
   $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis (1 + t232s1)}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pis (1 + t232s1)}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Eis = ais piseis == Sum[qis[t], {t, 1, i}];
```

```
ln[ ]:= Table[qns[t] =  $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns}\right) \left(\frac{pd[t]}{P[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs[t]}{pd[t]}\right) \left(\frac{pns}{zs[t]}\right)^{1-sigmas}$ , {t, 1, i}];
```

```
ln[ ]:= Ens = ans pnsens == Sum[qns[t], {t, 1, i}];
```

```
ln[ ]:= eqm = FindRoot[{Eda, Eia, Ena, Eds, Eis, Ens}, {pda, pda0},
  {pia, pia0}, {pna, pna0}, {pds, pds0}, {pis, pis0}, {pns, pns0}]
```

```
Out[ ]:= {pda → 0.992983, pia → 1.01947, pna → 0.99716, pds → 0.992512, pis → 1.01844, pns → 0.998077}
```

```
ln[ ]:= pda1 = pda /. eqm;
```

```
ln[ ]:= pia1 = pia /. eqm;
```

```
ln[ ]:= pna1 = pna /. eqm;
```

```

ln[ ]:= pds1 = pds /. eqm;

ln[ ]:= pis1 = pis /. eqm;

ln[ ]:= pns1 = pns /. eqm;

ln[ ]:= Table[za1[t] = If[adinp[t] + aiinp[t] + aninp[t] > 0,
  (pda11-sigmaa + bia[t] (pia1 (1 + t232a1))1-sigmaa + bna[t] pna11-sigmaa) $\frac{1}{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[zs1[t] = If[sdinp[t] + siinp[t] × sninp[t] > 0,
  (pds11-sigmas + bis[t] (pis1 (1 + t232s1))1-sigmas + bns[t] pns11-sigmas) $\frac{1}{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[pd1[t] = (w[[t]] + bs[t] × zs1[t] + ba[t] × za1[t]), {t, 1, i}];

ln[ ]:= Table[Pindex1[t] = (pd1[t]1-sigmad[[t]] + bi[t] pi[[t]]1-sigmad[[t]]) $\frac{1}{1-sigmad[[t]]}$ , {t, 1, i}];

ln[ ]:= Table[qd1[t] = k[t] Pindex1[t]sigmad[[t]]-1 pd1[t]-sigmad[[t]], {t, 1, i}];

ln[ ]:= qdatot1 = ada pda1eda;

ln[ ]:= qiatot1 = aia pia1eia;

ln[ ]:= qnatot1 = ana pna1ena;

ln[ ]:= qdstot1 = ads pds1eds;

ln[ ]:= qistot1 = ais pis1eis;

ln[ ]:= qnstot1 = ans pns1ens;

ln[ ]:= Table[qda1[t] = If[adinp[t] > 0,
  ( ( (k[t] × ba[t]) / pda1 ) ( pd1[t] / Pindex1[t] )1-sigmad[[t]] ( za1[t] / pd1[t] ) ( pda1 / za1[t] )1-sigmaa ), 0], {t, 1, i}];

ln[ ]:= Table[qia1[t] = If[aiinp[t] > 0, ( (k[t] × ba[t] × bia[t]) / pia1 (1 + t232a1) )
  ( pd1[t] / Pindex1[t] )1-sigmad[[t]] ( za1[t] / pd1[t] ) ( pia1 (1 + t232a1) / za1[t] )1-sigmaa, 0], {t, 1, i}];

ln[ ]:= Table[qna1[t] = If[aninp[t] > 0,
  ( (k[t] × ba[t] × bna[t]) / pna1 ) ( pd1[t] / Pindex1[t] )1-sigmad[[t]] ( za1[t] / pd1[t] ) ( pna1 / za1[t] )1-sigmaa, 0], {t, 1, i}];

ln[ ]:= Table[vda1[t] = If[adinp[t] > 0,
  pda1 * ( (k[t] × ba[t]) / pda1 ) ( pd1[t] / Pindex1[t] )1-sigmad[[t]] ( za1[t] / pd1[t] ) ( pda1 / za1[t] )1-sigmaa, 0], {t, 1, i}];

```

```

ln[ ]:= Table[via1[t] = If[aiinp[t] > 0, pia1 (1 + t232a1) *  $\left(\frac{k[t] \times ba[t] \times bia[t]}{pia1 (1 + t232a1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{za1[t]}{pd1[t]}\right) \left(\frac{pia1 (1 + t232a1)}{za1[t]}\right)^{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[vna1[t] = If[aninp[t] > 0, pna1 *  $\left(\frac{k[t] \times ba[t] \times bna[t]}{pna1}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{za1[t]}{pd1[t]}\right) \left(\frac{pna1}{za1[t]}\right)^{1-sigmaa}$ , 0], {t, 1, i}];

ln[ ]:= Table[qds1[t] = If[sdinp[t] > 0,
 $\left(\left(\frac{k[t] \times bs[t]}{pds1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pds1}{zs1[t]}\right)^{1-sigma}$ ), 0], {t, 1, i}];

ln[ ]:= Table[qis1[t] = If[siinp[t] > 0,  $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis1 (1 + t232s1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pis1 (1 + t232s1)}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[qns1[t] = If[sninp[t] > 0,
 $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pns1}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[vds1[t] = If[sdinp[t] > 0,
pds1 *  $\left(\left(\frac{k[t] \times bs[t]}{pds1}\right) \left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pds1}{zs1[t]}\right)^{1-sigma}$ ), 0], {t, 1, i}];

ln[ ]:= Table[vis1[t] = If[siinp[t] > 0, pis1 (1 + t232s1) *  $\left(\frac{k[t] \times bs[t] \times bis[t]}{pis1 (1 + t232s1)}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pis1 (1 + t232s1)}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

ln[ ]:= Table[vns1[t] = If[sninp[t] > 0, pns1 *  $\left(\frac{k[t] \times bs[t] \times bns[t]}{pns1}\right)$ 
 $\left(\frac{pd1[t]}{Pindex1[t]}\right)^{1-sigmad[[t]]} \left(\frac{zs1[t]}{pd1[t]}\right) \left(\frac{pns1}{zs1[t]}\right)^{1-sigma}$ , 0], {t, 1, i}];

```

7. Results

Aluminum

$$ln[\#] := 100 \left(\frac{pda0 - pda1}{pda1} \right)$$

Out[]:= 0.706656

$$ln[\#] := 100 \left(\frac{pia0 - pia1}{pia1} \right)$$

Out[]:= -1.90957

$$ln[\#] := 100 \frac{pia0 (1 + t232a0) - pia1 (1 + t232a1)}{pia1 (1 + t232a1)}$$

Out[]:= 7.89947

$$ln[\#] := 100 \left(\frac{pna0 - pna1}{pna1} \right)$$

Out[]:= 0.28478

$$ln[\#] := qdachange = 100 \left(\frac{qdatot0 - qdatot1}{qdatot1} \right)$$

Out[]:= 2.85673

$$ln[\#] := 100 \left(\frac{qiatot0 - qiatot1}{qiatot1} \right)$$

Out[]:= -31.9963

$$ln[\#] := 100 \left(\frac{qnatot0 - qnatot1}{qnatot1} \right)$$

Out[]:= 5.85236

Steel

$$ln[\#] := 100 \left(\frac{pds0 - pds1}{pds1} \right)$$

Out[]:= 0.754487

$$\text{In}[*]:= 100 \left(\frac{\text{pis0} - \text{pis1}}{\text{pis1}} \right)$$

Out[*]= -1.81087

$$\text{In}[*]:= 100 \frac{\text{pis0} (1 + \text{t232s0}) - \text{pis1} (1 + \text{t232s1})}{\text{pis1} (1 + \text{t232s1})}$$

Out[*]= 22.7364

$$\text{In}[*]:= 100 \left(\frac{\text{pns0} - \text{pns1}}{\text{pns1}} \right)$$

Out[*]= 0.192688

$$\text{In}[*]:= \text{qdschange} = 100 \left(\frac{\text{qdstot0} - \text{qdstot1}}{\text{qdstot1}} \right)$$

Out[*]= 1.8969

$$\text{In}[*]:= 100 \left(\frac{\text{qistot0} - \text{qistot1}}{\text{qistot1}} \right)$$

Out[*]= -23.9759

$$\text{In}[*]:= 100 \left(\frac{\text{qnstot0} - \text{qnstot1}}{\text{qnstot1}} \right)$$

Out[*]= 2.92964

Downstream

In[*]:= Inames(*naics codes*)

Out[*]= {2110, 312110, 3149, 3322, 3323, 3324, 3325, 3326, 3327, 3328, 3329,
3331, 3332, 3334, 3335, 3339, 3351, 3352, 3353, 3359, 336212, 336214,
336350, 336370, 336390, 336300, 336411, 3365, 3369, 3372, 3399, 3336, 3366}

$$\text{In}[*]:= \text{Table} \left[\text{pddchangex} [t] = 100 \left(\frac{\text{pd0}[t] - \text{pd1}[t]}{\text{pd1}[t]} \right), \{t, 1, i\} \right]$$

Out[*]= {0.03335, 0.20728, 0.0681025, 0.414865, 0.299717, 0.555181, 0.161144, 0.551412, 0.191944,
0.320037, 0.300397, 0.307754, 0.453637, 0.0917911, 0.184685, 0.232173, 0.133885,
0.1312, 0.125962, 0.119087, 0.24265, 0.0976961, 0.139418, 0.509113, 0.196901,
0.34009, 0.0448457, 0.276739, 0.26245, 0.15755, 0.0759604, 0.22521, 0.0508393}


```
In[ ]:= Table[100  $\left( \frac{\text{Pindex0}[t] - \text{Pindex1}[t]}{\text{Pindex1}[t]} \right)$ , {t, 1, i}]
```

```
Out[ ]:= {0.0227141, 0.1887, 0.0307058, 0.158667, 0.275523, 0.467169,  
0.0646122, 0.366341, 0.173137, 0.320037, 0.167437, 0.185292, 0.0742201,  
0.0591823, 0.114497, 0.123467, 0.0546875, 0.0379196, 0.0511168,  
0.0425836, 0.186755, 0.092209, 0.1091, 0.497399, 0.102583, 0.11177,  
0.0389042, 0.240294, 0.156823, 0.115857, 0.0209513, 0.0841869, 0.0464875}
```

```
In[ ]:= Table[qddchangex[t] = 100  $\left( \frac{\text{qd0}[t] - \text{qd1}[t]}{\text{qd1}[t]} \right)$ , {t, 1, i}]
```

```
Out[ ]:= {-0.0803099, -0.219249, -0.266948, -2.56228, -0.383931, -0.799119,  
-0.568227, -1.37127, -0.25189, -0.319016, -0.924406, -1.02881, -2.98053,  
-0.360569, -0.735456, -0.881246, -0.476053, -0.657395, -0.771971, -0.638304,  
-0.356616, -0.118573, -0.304789, -0.54155, -0.504332, -1.56638, -0.0566978,  
-0.497589, -0.907265, -0.293145, -0.392956, -1.24502, -0.0791119}
```

```
In[ ]:= Table[vddchangex[t] = (pd0[t] × qd0[t] - pd1[t] × qd1[t]), {t, 1, i}]
```

```
Out[ ]:= {-138508., -5354.65, -16475.9, -159450., -80701.7, -79036.,  
-24749.2, -62528.5, -35437.6, -3.72529 × 10-9, -321468., -440126.,  
-144043., -106190., -127394., -468912., -36496.3, -92414.9, -171521.,  
-159402., -9899.13, -3564.65, -56292.2, -10200.4, -135030., -132673.,  
-10190.7, -15978.4, -102265., -30964.3, -136123., -154233., -10285.2}
```

```
In[ ]:= Table[vdachange[t] = If[ vda1[t] > 0, vda0[t] - vda1[t], 0], {t, 1, i}];
```

```
In[ ]:= Table[vdschange[t] = If[ vds1[t] > 0, vds0[t] - vds1[t], 0], {t, 1, i}];
```

```
In[ ]:= domesticchange = Prepend[  
Table[{Inames[t], qddchangex[t], pddchangex[t],  $\frac{100 (\text{pd0}[t] \times \text{qd0}[t] - \text{pd1}[t] \times \text{qd1}[t])}{\text{pd1}[t] \times \text{qd1}[t]}$ ,  
vddchangex[t], vdachange[t], vdschange[t]}, {t, 1, i}],  
{"NAICS", "Percent change domestic quantity", "Percent change domestic price",  
"Percent change in value", "Change in value",  
"Change in domestic aluminum", "Change in domestic steel"}];
```

In[]:= Grid[domesticchange, Frame → All]

Out[]:=

NAICS	Percent change domestic quantity	Percent change domestic price	Percent change in value	Change in value	Change in domestic aluminum	Change in domestic steel
2110	-0.0803099	0.03335	-0.0469867	-138 508.	0	102 069.
312 110	-0.219249	0.20728	-0.0124233	-5354.65	200 877.	0
3149	-0.266948	0.0681025	-0.199028	-16 475.9	0	4765.78
3322	-2.56228	0.414865	-2.15805	-159 450.	6016.84	8177.65
3323	-0.383931	0.299717	-0.0853648	-80 701.7	74 337.2	213 523.
3324	-0.799119	0.555181	-0.248375	-79 036.	204 112.	53 562.3
3325	-0.568227	0.161144	-0.407999	-24 749.2	3517.84	6267.82
3326	-1.37127	0.551412	-0.827416	-62 528.5	1348.47	15 183.6
3327	-0.25189	0.191944	-0.0604294	-35 437.6	40 655.7	86 544.1
3328	-0.319016	0.320037	-1.50759 × 10 ⁻¹⁴	-3.72529 × 10 ⁻⁹	816.303	61 041.5
3329	-0.924406	0.300397	-0.626787	-321 468.	42 777.3	86 566.
3331	-1.02881	0.307754	-0.724221	-440 126.	-428.849	119 115.
3332	-2.98053	0.453637	-2.54042	-144 043.	1870.53	646.832
3334	-0.360569	0.0917911	-0.269109	-106 190.	21 752.	22 939.1
3335	-0.735456	0.184685	-0.552129	-127 394.	9854.93	26 727.6
3339	-0.881246	0.232173	-0.651119	-468 912.	29 901.5	104 217.
3351	-0.476053	0.133885	-0.342805	-36 496.3	9054.52	8328.9
3352	-0.657395	0.1312	-0.527058	-92 414.9	5768.35	15 576.4
3353	-0.771971	0.125962	-0.646981	-171 521.	5629.59	22 142.5
3359	-0.638304	0.119087	-0.519977	-159 402.	19 926.	20 794.1
336 212	-0.356616	0.24265	-0.114831	-9899.13	27 759.2	6570.06
336 214	-0.118573	0.0976961	-0.0209932	-3564.65	10 543.5	8911.94
336 350	-0.304789	0.139418	-0.165796	-56 292.2	15 060.2	30 582.2
336 370	-0.54155	0.509113	-0.0351942	-10 200.4	67 995.9	93 288.7
336 390	-0.504332	0.196901	-0.308424	-135 030.	39 615.5	54 411.
336 300	-1.56638	0.34009	-1.23162	-132 673.	12 448.	14 610.3
336 411	-0.0566978	0.0448457	-0.0118775	-10 190.7	1679.82	33 684.5
3365	-0.497589	0.276739	-0.222226	-15 978.4	108.608	15 738.4
3369	-0.907265	0.26245	-0.647196	-102 265.	21 419.2	20 803.1
3372	-0.293145	0.15755	-0.136057	-30 964.3	13 274.4	26 224.9
3399	-0.392956	0.0759604	-0.317294	-136 123.	10 338.1	23 037.8
3336	-1.24502	0.22521	-1.02261	-154 233.	646.952	16 837.7
3366	-0.0791119	0.0508393	-0.0283128	-10 285.2	18 870.2	8690.47